

Project Charter: Coffee Shop Sales Analysis

Project Title: Coffee Shop Sales Data Analysis

Project Objective:

The objective of this project is to analyze sales transactions of a coffee shop, identify trends, and generate insights that can support decision-making. The key goals include:

1. **Sales Performance Analysis:** Understanding revenue trends and sales patterns.
2. **Customer Behavior Analysis:** Identifying peak transaction times and popular products.
3. **Store Location Insights:** Analyzing performance across different store locations.
4. **Product Performance Evaluation:** Determining best-selling and underperforming products.
5. **Revenue Optimization:** Finding opportunities to increase revenue through data-driven decisions.

Project Scope:

- **Data Cleaning & Preparation:**
 - Handling missing values, duplicates, and data inconsistencies.
 - Ensuring correct data types (dates, numeric values, categories).
- **Exploratory Data Analysis (EDA):**
 - Identifying sales trends over time.
 - Analyzing peak hours and seasonal patterns.
 - Examining product-wise and store-wise revenue.
- **Descriptive & Statistical Analysis:**
 - Mean, median, and distribution of revenue.
 - Correlation between different variables (e.g., time of purchase vs. sales volume).
- **Visualization & Reporting:**
 - Creating dashboards in Power BI/Tableau.
 - Generating charts for revenue, product popularity, and store performance.

Data Overview:

- **Transaction ID:** Unique identifier for each sale.
- **Transaction Date & Time:** Timestamp of the sale.
- **Store ID & Location:** Identifies which store made the sale.
- **Product Details:** Category, type, and product name.
- **Unit Price & Quantity:** Determines revenue per transaction.
- **Total Revenue:** Revenue generated from each transaction.

Stakeholders:

- **Business Owners & Managers** – Use insights for strategic decisions.
- **Marketing Team** – Analyze product demand and promotions.
- **Operations Team** – Optimize inventory and staffing based on sales trends.

Tools & Technologies:

- **Power BI/Tableau** for visualization.
- **SQL/MySQL** for querying the dataset.
- **Python/Pandas** for data preprocessing and analysis.
- **Excel** for data validation and reporting.

Expected Deliverables:

1. **Cleaned & Processed Dataset** ready for analysis.
2. **Dashboard with Key Insights** on sales, peak times, and product trends.
3. **Detailed Report** summarizing findings, trends, and recommendations.
4. **Actionable Recommendations** for revenue optimization and business growth.

Timeline:

Phase	Tasks	Duration
Data Collection & Cleaning	Preprocessing and handling missing data	1 week
Exploratory Data Analysis	Visualizing trends and understanding patterns	1 week
Statistical Analysis	Identifying key insights and correlations	1 week
Reporting & Dashboarding	Creating visual reports and recommendations	1 week

Success Metrics:

- Improved understanding of sales trends.
- Identification of peak sales periods.
- Insights into customer preferences.
- Actionable strategies to optimize sales and revenue.